

JESSICA KALUIJI

DATA SCIENTIST / DATA ANALYST / BUSINESS INTELLIGENCE

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Portfolio | LinkedIn | GitHub

PROFESSIONAL PROFILE

Data scientist with a background in software development and data engineering. At First Quantum Minerals, I build ETL pipelines, prepare data from different systems, and develop reports and applications for teams across the business. I work mainly with Python, SQL Server and Power BI.

TECHNICAL SKILLS

Analysis & modelling: Python, pandas, NumPy, exploratory analysis, classification, feature engineering, model evaluation

Data & reporting: SQL Server, T-SQL, ETL, data validation, Power BI, DAX, Power Query, Excel

Development & automation: C#, .NET, EF Core, Django, React, REST APIs, Power Automate, Git

PROFESSIONAL EXPERIENCE

Graduate Development Programme, Data Science | First Quantum Minerals, Trident • Jul 2025 – Present

- Partner with managers and system owners across Safety, Mining, Process, HR and Training to define reporting measures, validation checks and usable outputs.
- Prepare and join operational and business data using SQL and Python; investigate duplicates, missing fields and inconsistent definitions.
- Develop Power BI reports with row-level access and automate recurring data preparation and alerts using Python and Power Automate.
- Present findings, validate reports with end users and adapt solutions as business processes and source systems change.

Platform Developer | Decarbonify AS • 2022 – 2025

- Developed ESG platform features using Django, React, Next.js, PostgreSQL and REST APIs.
- Designed database changes, integrated frontend and backend services, and tested improvements through Agile delivery.
- Translated stakeholder requirements into product features that improved data entry and reporting experiences.

Software Developer Intern | Decarbonify AS • 2021 – 2022

- Supported UI/UX research, wireframes, prototypes and design-to-development handover.
- Built responsive interfaces in React and Next.js and contributed to Django, REST API and PostgreSQL development.

SELECTED PROJECTS

Excess Hour and Recovery Monitoring | First Quantum Minerals

- Built SQL logic to pair clocking events, classify shifts and flag extended work periods and short recovery intervals.
- Delivered daily manager alerts and a Power BI report for prompt case review and historical trends.

Compliance and Competency Reporting | First Quantum Minerals

- Combined role requirements, learning records and reporting lines into department-level compliance and competency reporting.
- Defined expiry and assessment rules, resolved data-quality gaps and built proactive employee/supervisor alerts.

Unified Tasks and Approvals Dashboard | First Quantum Minerals

- Consolidated multiple workflow and learning systems into a governed SQL dataset and Power BI report with role-based access.
- Automated repeatable updates to provide one view of outstanding tasks and approvals.

Process Water Balance Application | First Quantum Minerals

- Developed a .NET and SQL Server application for validated monthly manual entries with controlled statuses.
- Aligned manual measurements with reconciled process data and prepared the reporting structure for Power BI.

Shift Guard | First Quantum Minerals

- Developing a predictive approach to identify shifts at higher risk of exceeding excessive hours from clocking, timing and recovery patterns.

- Planned classification evaluation and operational review; assessing wearable and fatigue-incident data as potential inputs.

ADDITIONAL ANALYTICS WORK

- Dispatch performance: defined queueing, delay, reassignment and unexpected-arrival measures for crew-level comparison.
- Succession and talent: prepared successor coverage, readiness and internal-movement measures for executive workforce reporting.
- Monthly data preparation: automated extraction and restructuring of a recurring operational workbook using Python and pandas.

EDUCATION

Master of Computer Applications (Artificial Intelligence) | Shoolini University • 2023 – 2025 | GPA 8.82

Bachelor of Computer Applications | Marwadi University • 2019 – 2022 | GPA 7.86

Grade 12 Certificate | Pestalozzi Education Centre • 2017 | 10 points

PUBLICATION AND RESEARCH

Published Springer book chapter (2026)

- Kaluiji, J., Chauhan, S., Sharma, A. K. & Wadhawan, S. “Can AI and IoT-Integrated Mobile Health Apps Effectively Improve Cognitive Health and Caregiver Support for Dementia?” In Cognitive Neuroscience and Artificial Intelligence, pp. 35–52.

Master’s thesis

- Design and Development of an AI Adaptive User Interface for Dementia Care — proposed and partially developed voice interaction, wake-word detection and ML-based command classification.

Unpublished research paper

- Cross-Domain Performance Evaluation of AI-Language Models — compared T5, MBart and MarianMT across domains and tasks using BLEU, ROUGE and METEOR.

References available on request.